

END SEMESTER ASSESSMENTS, SPRING 2024

Program: B.TECH COMPUTER SCIENCE & ENGINEERING / B.TECH COMPUTER SCIENCE & ENGINEERING (LATERAL)	
Subject Name: DIGITAL COMMUNICATION	
Subject Code: 3000013101 / 3000033101	Semester: IV
Duration: 2 Hours	Maximum Marks: 50

Note: Answer the following questions as directed.

Group A

(Answer All questions)

(Each question carries 1 mark)

1. Write the correct answer from the options given: (10 X 1)

- (i) In PAM, what property of the pulse is varied to carry information?
(a) Frequency (b) Phase
(c) Amplitude (d) Duration
- (ii) Which of the following demodulation techniques is commonly used for PAM?
(a) Envelope detection (b) Frequency demodulation
(c) Phase demodulation (d) Amplitude modulation
- (iii) In PCM, what is the purpose of quantization?
(a) To sample the analog signal (b) To encode the sampled values
(c) To reduce the bandwidth requirement (d) To demodulate the signal
- (iv) What is the function of the predictor in DPCM?
(a) To encode the quantized samples into binary code words (b) To sample the analog signal
(c) To estimate the value of the next sample (d) To decode the binary code words into quantized samples
- (v) What does a wider eye opening in an eye pattern indicate?
(a) Higher signal-to-noise ratio (b) Lower signal-to-noise ratio
(c) Greater signal distortion (d) Lower signal distortion

- (vi) If a bag contains 5 red balls and 3 blue balls, what is the probability of randomly selecting a red ball?
 (a) $5/8$ (b) $3/8$
 (c) $5/3$ (d) $3/5$
- (vii) What does the variance of a random variable measure?
 (a) How spread out the values are around the mean (b) The highest value
 (c) The lowest value (d) The most common value
- (viii) What is the area under the curve of a Gaussian distribution?
 (a) 1 (b) 0
 (c) Infinite (d) -1
- (ix) Which modulation technique is suitable for high-speed data transmission due to its efficient use of bandwidth?
 (a) ASK (b) FSK
 (c) PSK (d) None of the above
- (x) Which modulation technique is commonly used in RFID (Radio Frequency Identification) systems?
 (a) ASK (b) PSK
 (c) FSK (d) All of the above

Group B

(Answer any Four questions out of Six)
(Each question carries 5 marks)

(4 X 5)

2. What are the differences between natural and flat-top sampling?
3. Derive the S/N_Q for a sinusoidal signal with the maximum (minimum) amplitude $+A_m$ ($-A_m$).
4. Derive and draw the Power spectral density (PSD) of PNRZ signaling.
5. Explain μ -law companding.
6. What is Frequency Shift Keying (FSK) modulation? Explain the operation of the FSK modulator by block diagram representation. (2 + 3)
7. State the Nyquist sampling theorem. If the Quantization levels are 49, 125 calculate the no of bits needed in each of these PCM systems.

Group C

(Answer any Two questions out of Four)
(Each question carries 10 marks)

(2 X 10)

8. What is the significance of pdf in probability and random processes? For the given joint pdf $f(x, y) = xy^2 + \frac{x^2}{8}$, $0 < x < 2$; $0 < y < 1$. Find

(i) $P(X > 1)$

(ii) $P(Y > 1/2)$

(iii) $P(Y < 1/2 / X > 1)$

(2 + 8)

9. Explain intersymbol interference (ISI). Derive the Nyquist criterion for zero ISI. (4 + 6)

10. Discuss one major drawback of ideal sampling. An analog signal is to be transmitted over PCM system with an accuracy of $\pm 0.1\%$ (full scale). The signal has a bandwidth of 100 Hz and amplitude range from -10V to +10V. Calculate

(i) the number of bits per word

(ii) Signaling rate

(iii) Minimum transmission bandwidth

(2 + 4 + 2 + 2)

11. Describe delta modulation. Derive the bandwidth of PCM system. (6 + 4)